Course Objective and Outcome Form
Department of Electrical and Computer Engineering
School of Engineering and Physical Sciences
North South University, Bashundhara, Dhaka-1229, Bangladesh

1. **Course Number and Title: CSE 465 Pattern Recognition**
2. **Number of Credits: 03**
3. **Type: Elective**
4. **Prerequisites: CSE 373**
5. **Faculty Name: Dr. Nabeel Mohammed (NbM)**
6. **Room: SAC 917**
7. **Office Hours: TBA (Check my door)**
8. **Email: nabeel.mohammed@northsouth.edu**
9. **Contact Hours: Lectures – 3 Hours/week**
10. **Course Summary:**

    The course will cover topics such as supervised learning, unsupervised learning mainly in the context of neural networks. We will cover techniques, mainly popularised in the last few years, for neural network training and aim to give students an appreciation for the machine learning process involving this class of models.

11. **Resources**
    Text books:
    - **https://www.deeplearningbook.org/**
    - **http://neuralnetworksanddeeplearning.com/**

    (You will be expected to read a lot of academic papers)

12. **Topics:**
    a. **Supervised/Unsupervised Learning, Data, Dataset partitions, metrics (Lecture 1, 2)**
    b. **Regression, loss function, gradient descent (Lectures 3, 4)**

c. **Classification, entropy, KL divergence, cross entropy**
   (Lectures 5,6)
d. **Neural Networks**
   (Lecture 7,8)
e. **Convolutional Neural Networks**
   (Lectures 9 - 12)
f. **Recurrent Neural Networks, Transformers**
   (Lecture 13 - 16)
g. **Autoencoders**
   (Lecture 17)
h. **Generative Models**
   (Lecture 18 - 19)
i. **Reinforcement Learning**
   (Lectures 20 - 21)
j. **Many other possible topics if time permits**

13. Weightage Distribution among Assessment Tools

Assessment Tools	**Weightage (%)**
Project and related tasks	50
Final Exam	25
Midterm Exam	25
	100

14. Grading policy: As per NSU grading policy available in
http://www.northsouth.edu/academic/grading-policy.html

15. Course Policies:

a. Be good **and** follow university policies.